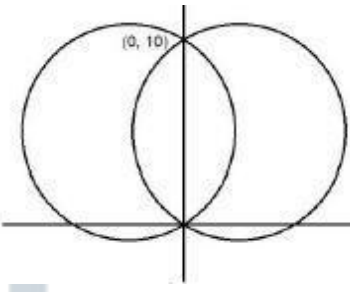
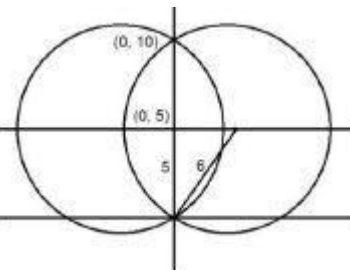


Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Selects correct answer	AO1.1b	B1	$\frac{2}{3}$
Total 1 mark			

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Produces a combined diagram showing circles intersecting at origin and (0, 10) or two separate diagrams. Allow reasonable 'hand drawn' circles which illustrate symmetry. Circles must cut the x axis again. Do not accept circles that go off the page.	AO2.2a	B1	
(b)	Deduces that y coordinate of centre is 5. (PI by any use of $(y - 5)$ in any circle equation or marked on diagram or seen as a y coordinate or used in Pythagoras)	AO2.2a	B1	
	Forms correct equation for x coordinate of centres using Pythagoras (PI)	AO1.1a	M1	$6^2 = 5^2 + a^2$ $a = +\sqrt{11} \text{ or } -\sqrt{11}$
	Obtains two correct circle equations (either form) Condone 3.3 or better provided $a = \sqrt{11}$ seen earlier	AO1.1b	A1	$(x - \sqrt{11})^2 + (y - 5)^2 = 6^2 = 36$ $(x + \sqrt{11})^2 + (y - 5)^2 = 6^2 = 36$ Or $x^2 + 2\sqrt{11}x + y^2 - 10y = 0$ $x^2 - 2\sqrt{11}x + y^2 - 10y = 0$
				Total 4 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses a technique which could lead to showing two lines are perpendicular. Obtains at least one correct distance (or distance ²) or gradient.	AO3.1a	M1	$AB^2 = (8 - 15)^2 + (17 - 10)^2$ $= 98$ $AC^2 = (8 - -2)^2 + (17 - -7)^2$ $= 676$ $CB^2 = (15 - -2)^2 + (10 - -7)^2$ $= 578$ $AB^2 + BC^2 = 98 + 578$ $= 676$ $= AC^2$ Angle ABC is a right angle.
	Obtains three correct distances (or distance ²) or two gradients. Lengths: $7\sqrt{2}, 17\sqrt{2}, 26$ Gradients: $AB = -\frac{7}{7}, BC = \frac{17}{17}$	AO1.1b	A1	
	Completes correct rigorous argument to show required result Uses Pythagoras OR Multiplies gradients to show product is -1 AND Writes a concluding statement.	AO2.1	R1	
(b)	Explains why AC is a diameter	AO2.4	E1	The angle subtended by a diameter is $90^\circ \therefore AC$ must be a diameter of the circle
(i)	Must reference angle subtended by diameter (condone "angle in a semi-circle") or give full explanation.			
(ii)	Deduces correct radius (or radius ²)	AO2.2a	B1	Radius $\frac{\sqrt{676}}{2} = 13$
	Obtains mid-point of diameter	AO1.1b	B1	Centre $\left(\frac{8-2}{2}, \frac{17-7}{2}\right) = (3, 5)$
	Uses $D(-8, -2)$ to find the distance or (distance ²) from their centre OE	AO1.1a	M1	Distance from centre to D $(3 - -8)^2 + (5 - -2)^2 = 11^2 + 7^2$ $= 170$
	Completes rigorous argument by comparing $\sqrt{170} > 13$ (or $170 > 169$) to show that D lies outside the circle	AO2.1	R1	> 169 So D lies outside the circle.
				Total 8 marks

Q4.

Marking Instructions	AO	Marks	Typical Solution
Identifies and clearly defines consistent variables for length and width. Can be shown on diagram.	AO3.1b	B1	Width of rectangle = $2x$ Length of rectangle = $2y$
Models the area of rectangle with an expression of the correct dimensions	AO3.3	M1	$A = 4xy$
Eliminates either variable to form a model for the area in one variable.	AO1.1a	M1	$x^2 + y^2 = 16$
Obtains a correct equation to model the area in one variable	AO1.1b	A1	$A = 4x\sqrt{16 - x^2}$
Differentiates their expression for area. Condone one error	AO3.4	M1	$\frac{dA}{dx} = 4\sqrt{16 - x^2} - \frac{4x^2}{\sqrt{16 - x^2}}$
Explains that their derivative equals zero for a maximum or stationary point.	AO2.4	E1	$\frac{dA}{dx} = \frac{64 - 8x^2}{\sqrt{16 - x^2}}$ For maximum point $\frac{dA}{dx} = 0$
Equates area derivative to zero and obtains correct value for either variable. CAO	AO1.1b	A1	$\frac{64 - 8x^2}{\sqrt{16 - x^2}} = 0$ $x = 2\sqrt{2}$
Completes a gradient test or uses second derivative of their area function to determine nature of stationary point	AO1.1a	M1	When $x = 2.8$, $\frac{dA}{dx} = 0.448$ When $x = 2.9$, $\frac{dA}{dx} = -1.191$ Therefore maximum
Deduces that the area is a maximum at $x = 2\sqrt{2}$ or $\theta = \frac{\pi}{4}$ Values need not be exact	AO2.2a	R1	The maximum area is 32 sq in
Obtains maximum area with correct units AWRT 32	AO3.2a	B1	
			Total 10 marks

Q5.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	9π
Total 1 mark			

Q6.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States correct radius CAO	AO1.2	B1	Radius = $\sqrt{5}$
(i)				
(ii)	States correct centre CAO	AO1.2	B1	C is (7, -2)
(b)	Finds gradient of the line through the points <i>P</i> and 'their' <i>C</i> (as found in part (a)) Condone one sign error	AO3.1a	M1	Gradient $CP = \frac{-1 - (-2)}{5 - 7} = -\frac{1}{2}$
	Correct tangent gradient obtained from 'their' CP gradient	AO3.1a	M1	So tangent gradient = 2
	Uses a correct form for the equation of a straight line with correct coordinates of <i>P</i> and 'their' tangent gradient	AO1.1a	M1	$y - (-1) = 2(x - 5)$
	States correct final answer in required form ($y = mx + c$) FT from 'their' <i>C</i> found in part (a)	AO1.1b	A1F	$y = 2x - 11$
(c)	Identifies QTC as a right-angled triangle PI	AO3.1a	M1	QTC is a right-angled triangle so we can use Pythagoras
	Finds QC or QC^2 FT 'their' <i>C</i> found in part (a)	AO1.1b	B1F	$QC^2 = (7 - 3)^2 + (-2 - 3)^2$
	Uses Pythagoras' theorem correctly for 'their' triangle	AO1.1a	M1	$4^2 + 5^2 = (\sqrt{5})^2 + QT^2$
	Correct evaluation of length of QT	AO1.1b	A1F	$QT^2 = 36$ so $QT = 6$

FT 'their' QC and 'their' radius found in part (a)		
Total 10 marks		

Q7.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Completes the square twice or applies standard formula	AO1.1a	M1	$(x + 4)^2 + (y - 6)^2 - 16 - 36 = 12$ $(x + 4)^2 + (y - 6)^2 = 64$
	Obtains correct equation	AO1.1b	A1	Centre $(-4, 6)$
	Obtains correct radius and correct coordinates of C Follow through 'their' equation	AO1.1b	A1F	Radius = 8
(b)	Demonstrates a method to find the length OP or OQ (or their squares), or the coordinates P or Q using 'their' values from part (a)	AO3.1a	M1	$OC^2 = 4^2 + 6^2 = 52$ $OP^2 = r^2 - OC^2$ $= 64 - 52 = 12$ $PQ = 2OP$ $2\sqrt{12} = 4\sqrt{3}$
	Uses a circle property that may lead to a solution, eg radius and chord meet at right-angles (evidence for this could be the use of Pythagoras or perpendicular gradients)	AO3.1a	M1	
	Finds OP or OQ or coordinates of P or Q CAO	AO1.1b	A1	
	Obtains length of PQ Follow through from 'their' coordinates of P and Q (Does not need to be in the required form)	AO1.1b	A1F	
	Sets out a well-constructed mathematical argument, using precise statements and correct use of symbols throughout to show the correct required result in required form Only award if they have a completely correct	AO2.1	R1	

solution, which is clear, easy to follow and contains no slips			
Total 8 marks			

Q8.

(a) $x = 0 \Rightarrow y^2 - 4y - 12 = 0$

sub $x = 0$ & correct quadratic in y

or $(y - 2)^2 = 16$ or $(y - 2)^2 - 16 = 0$

M1

$(y - 6)(y + 2) = 0$

correct factors or formula as far as $\frac{4 \pm \sqrt{64}}{2}$

or $y - 2 = \pm \sqrt{16}$

A1

$\Rightarrow y = -2, 6$

condone $(0, -2)$ & $(0, 6)$

A1

3

(b) $(x + 3)^2 - 9 + (y - 2)^2 - 4 (=12)$

correct sum of square terms and attempt to complete squares (or multiply out) PI by next line

M1

$(r^2 =) 9 + 4 + 12$

$(r^2 =) 25$ seen on RHS

A1

$(\Rightarrow r =) 5$

$r = \sqrt{25}$ or $r = \pm 5$ scores A0

A1

3

(c) (i) $(CP^2 =) (2 - 3)^2 + (5 - 2)^2$

condone one sign slip within one bracket

M1

$\Rightarrow (CP =) \sqrt{34}$

$$n = 34$$

A1

2

(ii) $PQ^2 = CP^2 - r^2 = 34 - 25$

Pythagoras used correctly with values FT "their" r and CP

M1

$(\Rightarrow PQ =) \quad 3$

A1

2

[10]

Q9.

(a) $(x - 5)^2 + (y + 7)^2$

one term correct

M1

both terms correct and added

A1

$(x - 5)^2 + (y + 7)^2 = 49$

must see 49 not just 7^2

condone $(x - 5)^2 + (y - - 7)^2 = 49$

A1cao

3

(b) (i) (Centre is) $(5, -7)$

correct or FT their a and b

B1✓

1

(ii) Radius = 7

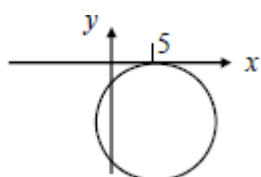
*condone 49 but **not** ± 7 or $\pm \sqrt{49}$*

correct or FT their $\sqrt{k}$ provided $k > 0$

B1✓

1

(c) (i)



*freehand circle with centre in correct quadrant or FT
from their (b)(i) must have both axes shown clearly*

M1

*correct position cutting negative y-axis twice and
touching x-axis at $x = 5$*

*5 must be marked on x-axis or centre clearly marked
as $(5, -7)$*

must have correct centre and radius in (b)

A1

2

(ii) $x = 5$

B1

$$y = -14$$

$$(5, -14)$$

B1

2

(d) Translation

and no other transformation

E1

through $\begin{bmatrix} 6 \\ * \end{bmatrix}$

M1

$$\begin{bmatrix} 6 \\ -7 \end{bmatrix}$$

*both components correct for A1; may describe in
words or use a column vector*

A1cso

3

[12]

Q10.

(a) $(x + 7)^2 + (y - 5)^2$

one term correct ; condone $(x - -7)^2$

M1

*both terms correct with squares
and plus sign between terms*

A1

$$(x + 7)^2 + (y - 5)^2 = 5^2$$

condone 25 for 5²

A1cao

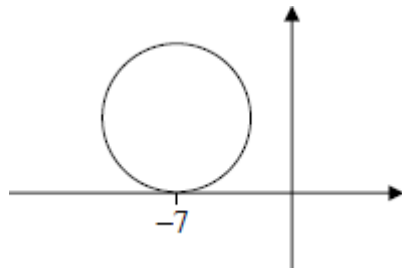
- (b) (i) $C(-7, 5)$
correct or FT 'their' circle equation

B1✓

- (ii) $r = 5$
correct or FT 'their' $r^2 > 0$
condone $\sqrt{25}$ etc but not $\pm \sqrt{25}$

B1✓

- (c) must draw axes



freehand circle with C correct or FT 'their C' for quadrant of centre

M1

circle touching x-axis at -7 with -7 marked (need not show 5 on y-axis) but circle must not touch y-axis

A1

- (d) (i) $x^2 + (kx + 6)^2 + 14x - 10(kx + 6) + 49 = 0$
clear attempt to sub $y = kx + 6$ into original or 'their' circle equation ...

$$x^2 + k^2x^2 + 12kx + 36 + 14x - 10kx - 60 + 49 = 0$$

...and attempt to multiply out

M1

$$(1 + k^2)x^2 + 2kx + 14x + 25 = 0$$

$$\Rightarrow (k^2 + 1)x^2 + 2(k + 7)x + 25 = 0$$

AG *condone $x^2(1 + k^2) + 2x(7 + k) + \dots$ etc*

A1cso

- (ii) Equal roots ' $b^2 - 4ac = 0$ '

allow statement alone if discriminant in terms of k attempted

B1

$$[2(k+7)]^2 - 4 \times 25(k^2 + 1)$$

discriminant (condone one slip)

M1

$$4\{k^2 + 14k + 49 - 25k^2 - 25\} = 0$$

$$-24k^2 + 14k + 24 = 0$$

$$\Rightarrow 12k^2 - 7k - 12 = 0$$

AG all working correct
but = 0 must appear before last line

A1

3

(iii) $(4k+3)(3k-4)$

correct factors or correct use of formula as far as

$$k = \frac{7 \pm \sqrt{49 + 576}}{24}$$

M1

$$\Rightarrow k = -\frac{3}{4}, k = \frac{4}{3} \quad \text{OE}$$

are values of k for which line is a tangent

A1

2

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[14]

Q11.

(a) $(x-5)^2 + (y-8)^2$

B1

$$= 25$$

condone 5^2

B1

2

(b) (i) $(2-5)^2 + (12-8)^2$
or $AC^2 = 3^2 + 4^2$

$$= 9 + 16 = 25$$

hence $AC = 5$; (also radius = 5)

$\Rightarrow A$ lies on circle

CSO

($\Rightarrow$ radius = AC) $\Rightarrow A$ lies on circle

B1

(must have concluding statement and circle equation correct if using equation)

(must have concluding statement & RHS of circle equation correct or $r = 5$ stated if Pythagoras is used)

1

(ii) grad = $\frac{4}{3}$

B1

Gradient of tangent is $\frac{3}{4}$
 FT their $-1/\text{grad } AC$

B1 ✓

$y - 12 = \text{'their tangent grad'} (x - 2)$
 or $y = \text{'their tangent grad'} x + c$
 & attempt to find c using $x = 2, y = 12$

M1

$y - 12 = \frac{3}{4}(x - 2)$ or $y = \frac{3}{4}x + \frac{21}{2}$ etc
 correct equation in any form

A1

$3x - 4y + 42 = 0$

CSO must have integer coefficients with all terms on one side of equation accept $0 = 8y - 6x - 84$ etc

A1

5

(c) (i) $(CM^2 =) (7 - 5)^2 + (12 - 8)^2$
 or $(CM^2 =) 20$

M1

($\Rightarrow CM = \sqrt{20}$) $\Rightarrow (CM =) 2\sqrt{5}$

A1

2

(ii) $PM^2 = PC^2 - CM^2 = 25 - 20$

Pythagoras used correctly eg $d^2 + (2\sqrt{5})^2 = 5^2$

M1

$\Rightarrow PM = \sqrt{5}$

A1

Area $\Delta PCQ = \sqrt{5} \times 2\sqrt{5} = 10$
CSO

A1

3

[13]

Q12.

(a) (i) $(x + 3)^2 + (y - 1)^2$
condone $(x - -3)^2$

$= 13$

condone $(\sqrt{13})^2$

B1

B1

2

(ii) $x^2 + 6x + 9 + y^2 - 2y + 1$

*attempt to multiply out both of 'their' brackets;
must have x and y terms*

M1

$x^2 + y^2 + 6x - 2y$

both $m = 6$ and $n = -2$

A1

$-3 = 0$

All correct, $p = -3$ and $\dots = 0$

A1

3

(b) $x = 0 \Rightarrow y^2 - 2y - 3 = 0 \Rightarrow (y - 3)(y + 1) = 0$

*putting $x = 0$ PI
 and attempt to solve or factorise*

M1

$y = 3, y = -1$

A1

$\Rightarrow \text{Distance } AB = 3 + 1 = 4$

OR Pythagoras $d^2 = 13 - 3^2$ M1

$d = 2$ A1

$\text{distance} = 2 \times 2 = 4$ A1

A1cso

3

(c) (i) $(-5 + 3)^2 + (-2 - 1)^2 = 4 + 9 = 13$

Substitution $x = -5, y = -2$ into any correct circle equation

$\Rightarrow D$ lies on circle

*convincing verification **plus statement***

B1

1

(ii) $\text{grad } CD = \frac{1+2}{-3+5}$

condone one sign slip

M1

$= \frac{3}{2}$ (or 1.5)

not $\frac{-3}{-2}$

A1

2

(iii) Perpendicular gradient $= -\frac{2}{3}$

ft their grad CD or $m_1 m_2 = -1$ stated

M1

Tangent has equation $y + 2 = -\frac{2}{3}(x + 5)$

any form of correct equation eg $2x + 3y + 16 = 0$

$$y = -\frac{2}{3}x + c, \quad c = -\frac{16}{3}$$

A1

2

[13]

Q13.

(a) $(x - 3)^2 + (y + 8)^2$

accept $(y - -8)^2$

B1

$= 100$

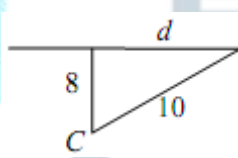
condone $RHS = 10^2$ or $k = 10^2$

B1

2

(b) $y = 0 \Rightarrow$ 'their' $(x - a)^2 + b^2 = k$

Alternative



M1

$(x - 3)^2 = 36$ or $x^2 - 6x - 27 (= 0)$ (PI)

A1

$\Rightarrow x = -3, 9$

$(d^2 =) 10^2 - 8^2$ M1

$d^2 = 36$ A1 or $d = 6$

$\Rightarrow x = -3, 9$ A1

A1

3

(c) Line CA has gradient $-\frac{2}{5}$

M1

$(y + 8) = -\frac{2}{5}(x - 3)$
CA has equation
any form of correct equation
eg $y = -\frac{2}{5}x + c, \quad c = -\frac{34}{5}$

A1

$2x + 5y + 34 = 0$
integer coefficients – all terms on 1 side

A1cso

3

(d) (i) their $(x - 3)^2 + (2x + 1 + 8)^2$ or

$x^2 + (2x + 1)^2 - 6x + 16(2x + 1) (+ 73)$
substituting $y = 2x + 1$ correctly into LHS
of “their” circle equation and attempt to
expand in terms of x only

M1

$x^2 - 6x + 9 + 4x^2 + 36x + 81 = 100$
 or $x^2 + 4x^2 + 4x + 1 - 6x + 32x + 16 + 73 = 100$
any correct equation (with brackets expanded)

A1

$\Rightarrow 5x^2 + 30x - 10 = 0$
must see this line or equivalent

$$\Rightarrow x^2 + 6x - 2 = 0$$

AG; all algebra must be correct

A1cso

3

(ii) $(x + 3)^2 = 11$
or correct use of formula

must get as far as $x = \frac{-6 \pm \sqrt{44}}{2}$

M1

$x = -3 \pm \sqrt{11}$
exactly this

A1cso

2

[13]

Q14.

(a) (i) $x = \pm 2$ or $y = \pm 6$ or $(x - 2)^2 + (y + 6)^2$

M1

$C \quad 2, -6$
correct

A1

2

(ii) $r^2 = 4 + 36 - 15$
(RHS =) their $(-2)^2 + \text{their } (6)^2 - 15$

M1

$\Rightarrow r = 5$

Not ± 5 or $\sqrt{25}$

A1

2

(b) explaining why $|y_c| > r$; $6 > 5$
Comparison of y_c and r , eg $-6 + 5 = -1$
or indicated on diagram

E1

full convincing argument, but must have
correct y_c and r

Eg "highest point is at $y = -1$ " scores E2
E1: showing no real solutions when $y = 0$
+E1 stating centre or any point below
 x -axis

E1

2

(c) (i) $PC^2 = 5 - 2^2 + k + 6^2$
ft their C cords
 $= 9 + k^2 + 12k + 36$
and attempt to multiply out

M1

$$PC^2 = k^2 + 12k + 45$$

AG CSO (must see $PC^2 =$ at least once)

A1

2

$$(ii) \quad \left. \begin{aligned} PC > r &\Rightarrow PC^2 > 25 \\ \Rightarrow k^2 + 12k + 20 &> 0 \end{aligned} \right\}$$

$$\left. \begin{aligned} k^2 + 12k + 45 &> 25 \\ \Rightarrow k^2 + 12k + 20 &> 0 \end{aligned} \right\} \text{AG Condone}$$

B1

1

$$(iii) \quad k + 2 \quad k + 10$$

Correct factors or correct use of formula

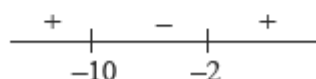
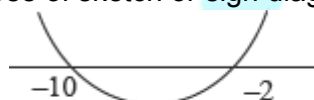
M1

$k = -2, k = -10$ are critical values

May score M1, A1 for correct critical values seen as part of incorrect final answer with or without working.

A1

Use of sketch or sign diagram:



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If previous A1 earned, sign diagram or sketch must be correct for M1, otherwise M1 may be earned for an attempt at the sketch or sign diagram using their critical values.

M1

$$\Rightarrow k > -2, k < -10$$

$k \geq -2, k \leq -10$ loses final A mark

A1

Condone $k > -2$ OR $k < -10$ for full marks but not AND instead of OR
Take final line as their answer

Answer only of $k > -2$, $k > -10$ etc
scores M1, A1, M0 since the critical
values are evident.

Answer only of $k > 2$, $k < -10$ etc scores
M0, M0 since the critical values are not
both correct.

4

[13]

Q15.

(a) $(x + 5)^2 + (y - 6)^2 = 5^2$

One term correct LHS

M1

LHS all correct

A1

RHS correct: condone = 25

B1

3

(b) (i) sub $x = -2$, $y = 2$ into circle equation

Circle equation must be correct

$3^2 + (-4)^2 = 25 \Rightarrow$ lies on circle

Must have concluding statement

B1

1

(ii) Grad $PC = -\frac{4}{3}$

Condone $\frac{4}{-3}$

B1

Normal to circle has equation

$y - 6 = \text{'their gradient } PC' (x + 5)$

or $y - 2 = \text{'their gradient } PC' (x + 2)$

M0 if tangent attempted or incorrect
coordinates used

M1

$$y - 6 = -\frac{4}{3}(x + 5)$$

$$\text{or } y - 2 = -\frac{4}{3}(x + 2)$$

Any correct form eg $4x + 3y + 2 = 0$

$$y = -\frac{4}{3}x + c, \quad c = -\frac{2}{3}$$

A1cso

3

$$(iii) \quad PM = \frac{1}{2} \times \text{radius}$$

M1

$$= 2.5$$

A1cso

$$PO = \sqrt{8}$$

B1

P is closer to the point M

E1 cso

Alternative 1

Attempt at $M\left(-\frac{7}{2}, 4\right)$ with at least one correct coordinate **and** PM^2 attempted

(M1)

$$PM^2 = \frac{9}{4} + 4 = \frac{25}{4}$$

(A!cso)

$$PO^2 = 4 + 4 = 8$$

(B1)

Statement following correct values

(E1cso)

Alternative 2

Attempt at $M\left(-\frac{7}{2}, 4\right)$ with at least one

correct coordinate **and** attempt at vectors
or difference of coordinates

(M1)

$$\overrightarrow{PM} = \begin{pmatrix} -1.5 \\ 2 \end{pmatrix} \quad \text{OE}$$

(A1cso)

P is closer to the point M

(E1cso)

Components of their $\overrightarrow{PM}$ and $\overrightarrow{OP}$
considered – **totally independent** of M1

(E1)

4

[11]

Q16.

(a) (i) $(x - 3)^2 + (y + 5)^2$

One term correct

B1

LHS correct with + and squares

B1

$$= 25 - 9 + 9 = 25 \quad (= 5^2)$$

Condone RHS = 25

B1

3

(b) (i) $C(3, -5)$

B1ft

(ii) Radius = 5

Correct or ft their RHS provided > 0

B1ft

2

(c) (i) $(7 - 3)^2 + (-2 + 5)^2 = 16 + 9 = 25$

Or sub'n of (7, -2) in original equation

$\Rightarrow D$ lies on circle

Must see statement

$$7^2 + (-2)^2 - 42 - 20 + 9 = 0$$

Or sub $x = 7$ into eqn & showing $y = -2$ etc

B1

1

(ii) Attempt at gradient of CD as normal grad $CD =$

$$\frac{-2 - (-5)}{7 - 3} = \frac{3}{4}$$

withhold if subsequently uses $m_1 m_2 = -1$

$$\frac{\Delta y}{\Delta x}$$

(condone one slip) FT their centre C

M1

$$y + 2 = \frac{3}{4}(x - 7) \text{ or } y + 5 = \frac{3}{4}(x - 3)$$

Correct equation in any form $y = \frac{3}{4}x - \frac{29}{4}$

A1

$$\Rightarrow 3x - 4y = 29$$

CSO **Integer** coefficients

Condone $4y - 3x + 29 = 0$ etc

A1

3

(d) (i) $y = kx$ sub'd into original circle equation

or using their completed square form and multiplying out

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M1

$$x^2 + (kx)^2 - 6x + 10kx + 9 = 0$$

$$\Rightarrow (k^2 + 1)x^2 + 2(5k - 3)x + 9 = 0 \quad \mathbf{AG}$$

CSO

must see at least previous line for A1

any error such as $kx^2 = \dots = k^2 x^2$ gets A0

A1

2

(ii) $4(5k - 3)^2 - 36(k^2 + 1)$

Discriminant in k (can be seen in quad formula)

Condone one slip

M1

$$= 64k^2 - 120k$$

$$\text{or } 8k^2 - 15k = 0 \quad \text{OE}$$

A1

Equal roots: $4(5k - 3)^2 - 36(k^2 + 1) = 0$
 $b^2 - 4ac = 0$ clearly stated or evident by
 an equation in k with at most 2 slips.

B1

$$8k^2 - 15k = 0$$

Attempt to solve **their** quadratic or linear
 equation if k has been cancelled

m1

$$\Rightarrow k = 0, k = \frac{15}{8}$$

OE but must have $k = 0$
 If “= 0” is not seen but correct values of k
 are found, candidate will lose B1 mark but
 may earn all other marks

A1

5

- (iii) (Line is a) **tangent** (to the circle)
 Line **touches** circle at one point

E1

1

[17]

Q17.

- (a) (i) $C(5, -12)$

B1

1

- (ii) Radius = 13 (or $\sqrt{169}$)
 $\pm \sqrt{169}$ or ± 13 as final answer scores B0

B1

1

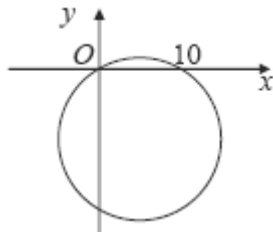
- (b) (i) $(-5)^2 + 12^2$ or $25 + 144$
 $= 169 \Rightarrow$ circle passes through O
 Correct arithmetic plus statement

Eg "O lies on circle", "as required" etc

B1

1

(ii) Sketch



Freehand circle through origin and cutting positive x-axis with centre in 4th quadrant
Condone value 10 missing or incorrect

B1

$$25 + (p + 12)^2 = 169$$

Or doubling their y_C -coordinate

M1

$$(p + 12) = \pm 12 \quad p = -24$$

Condone use of y instead of p
SC B2 for correct value of p stated or marked on diagram

A1

3

(c) (i) $\text{grad } AC = \frac{-12 + 7}{5 + 7}$
correct expression, but ft their C

M1

$$= -\frac{5}{12}$$

Condone $\frac{5}{-12}$

A1

2

(ii) $\text{grad tangent} = \frac{12}{5}$

$$\frac{-1}{\text{their grad } AC}$$

B1ft

$$y + 7 = \frac{12}{5}(x + 7)$$

ft “their $\frac{12}{5}$ ” must be **tangent** and not AC

M1

$$\Rightarrow 12x - 5y + 49 = 0$$

OE with **integer** coefficients with all terms on one side of the equation

A1

3

[11]

Q18.

(a) $x^2 + (y - 5)^2$

$$b = 5$$

B1

$$\text{RHS} = 5$$

$$k = 5$$

B1

2

- (b) (i) Centre (0, 5) *FT their b from part (a)*

B1ft

1

(ii) Radius = $\sqrt{5}$

FT their k from part (a); RHS must be > 0

B1ft

1

(c) (i) $x^2 + 4x^2 - 20x + 20 = 0$

May substitute into original or “their (a)”

M1

$$\Rightarrow x^2 - 4x + 4 = 0$$

CSO; AG

A1

2

(ii) $(x - 2)^2 = 0$ or $x = 2$

M1

Repeated root implies tangent

Or $b^2 - 4ac$ shown = 0 plus statement

E1

Point of contact is $P(2, 4)$

A1

3

(d) $(CQ^2 =) 1^2 + 1^2$

FT their C

M1

$$\sqrt{2} < \sqrt{5} \Rightarrow Q \text{ lies inside circle}$$

CQ or CQ^2 OE must appear for A1

A1CSO

2

[11]

Q19.

(a) $(x - 8)^2 + (y - 13)^2$

Exactly this with + and squares

B1

$$= 13^2$$

Condone 169

B1

2

(b) (i) $\text{grad } PC = \frac{12}{5}$

Must simplify $\frac{-12}{-5}$

B1

(ii) grad of tangent $= \frac{-1}{\text{grad } PC} = -\frac{5}{12}$
Condone $-\frac{1}{2.4}$ etc

B1ft

tangent has equation $y - 1 = -\frac{5}{12}(x - 3)$
ft gradient but M0 if using grad PC

M1

Correct – but not in required final form

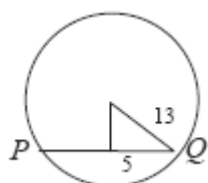
A1

$5x + 12y = 27$ OE
MUST have integer coefficients

A1

4

(iii)



half chord = 5
Seen or stated

B1

$d^2 = (\text{their } r)^2 - 5^2$
 (provided $r > 5$)
Pythagoras used correctly $d^2 = 13^2 - 5^2$

M1

Distance = 12
 CSO

A1

3

[10]

Q20.

(a) $(x + 1)^2 + (y - 6)^2$

B1 for one term correct or missing + sign

B2

$(1 + 36 - 12 = 25)$ RHS = 5^2

Condone 25

B1

3

(b) (i) Centre $(-1, 6)$

FT their a and b from part (a) or correct

B1 ✓

1

(ii) Radius = 5

FT their r from part (a) RHS must be > 0

B1 ✓

1

(c) Attempt to solve "their" $x^2 + 2x + 12 = 0$

Or comparing "their" $y_c = 6$ and their $r = 5$

M1

(all working correct) so no real roots
or statement that does not intersect

may use a diagram with values shown

$\begin{cases} r < y_c \text{ so does not intersect} \\ \text{condone } \pm 1 \text{ or } \pm 6 \text{ in centre for A1} \end{cases}$

A1

2

(d) (i) $(4 - x)^2 = 16 - 8x + x^2$

Or $(-2 - x)^2 = 4 + 4x + x^2$

B1

$x^2 + (4 - x)^2 + 2x - 12(4 - x) + 12 = 0$

or $(x + 1)^2 + (-2 - x)^2 = 25$

*Sub $y = 4 - x$ in circle eqn (condone slip)
or "their" circle equation*

M1

$$\Rightarrow 2x^2 + 6x - 20 = 0 \quad \Rightarrow x^2 + 3x - 10 = 0$$

AG CSO (must have = 0)

A1

3

(ii) $(x + 5)(x - 2) = 0 \quad \Rightarrow x = -5, x = 2$

Correct factors or unsimplified solution to quadratic (give credit if factorised in part (i))

M1

Q has coordinates $(-5, 9)$

SC2 if Q correct. Allow $x = -5 \quad y = 9$

A1

2

(iii) Mid point of 'their' $(-5, 9)$ and $(2, 2)$

*Arithmetic mean of **either** x or y coords*

M1

$$\left(-1\frac{1}{2}, 5\frac{1}{2}\right)$$

Must follow from correct value in (ii)

A1

2

[14]

Q21.

(a) (i) Centre $(-3, 2)$

± 3 or ± 2

M1

correct

A1

2

(ii) Radius = 5

accept $\sqrt{25}$ but not $\pm \sqrt{25}$

B1

1

(b) (i) $3^2 + (-4)^2 = 9 + 16 = 25$

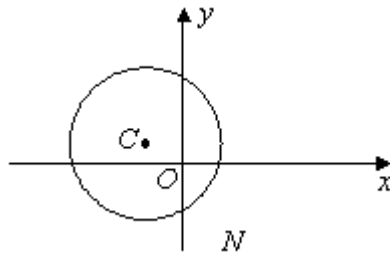
$\Rightarrow N$ lies on circle

must have $9 + 16 = 25$ or a statement

B1

1

(ii)



*must draw axes;
ft their centre in correct quadrant*

M1

correct (reasonable freehand circle enclosing origin)

A1

2

(iii) Attempt at gradient of CN
withhold if subsequently finds tangent

M1

$$\text{grad } CN = -\frac{4}{3}$$

CSO

A1

$$y = -\frac{4}{3}x - 2$$

(or equivalent)

ft their grad CN

A1 ✓

3

(c) (i) $P(2, 6)$ Hence $PC^2 = 5^2 + 4^2$
"their" PC^2

M1

$$\Rightarrow PC = \sqrt{41}$$

A1

2

(ii) Use of Pythagoras correctly

M1

$$PT^2 = PC^2 - r^2 = 41 - 25,$$

where t is a point of contact of tangent

ft their PC^2 and r^2

A1 ✓

$$\Rightarrow PT = 4$$

Alternative

sketch with vertical tangent M1

showing that tangent touches circle at

point (2, 2) A1

hence $PT = 4$ A1

A1

3

[14]

Q22.

(a) $(x - 4)^2 + (y + 3)^2$

B1 for one term correct

B2

$(11 + 16 + 9 = 36)$

RHS = 6^2

Condone 36

B1

3

(b) (i) Centre $(4, -3)$

Ft their a and b from part (a)

B1ft

(ii) Radius = 6

Ft their r from part (a)

B1ft

(c) (i) $CO^2 = (-4)^2 + 3^2$

Accept + or - with numbers but must add

M1

$CO = 5$

Full marks for answer only

A1ft

2

- (ii) Considering CO and radius

M1

$CO < r \Rightarrow O$ is **inside** the circle

Ft outside circle when 'their CO ' $> r$

or on the circle when 'their CO ' $= r$

SC B1ft if no explanation given

A1ft

2

[9]

Q23.

- (a) (i) $(x - 2)^2$

Attempt to complete square for x

M1

centre has x -coordinate = 2

M1 implied if value correct or -2

A1

and y -coordinate = 0

Centre (2,0)

B1

3

- (ii) RHS = 18

Withhold if circle equation RHS incorrect

B1

Radius = $\sqrt{18}$

Square root of RHS of equation (if > 0)

M1

Radius = $3\sqrt{2}$

A1

3

- (b) Perpendicular bisects chord so need to use
Length of 4

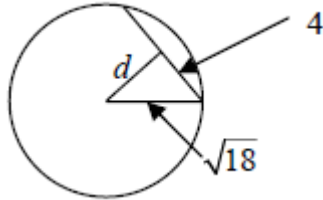
B1

$$d^2 = (\text{radius})^2 - 4^2$$

M1

$$d^2 = 18 - 16 \text{ so perpendicular distance} = \sqrt{2}$$

A1



3

(c) (i) $x^2 + (2k - x)^2 - 4x - 14 = 0$

M1

$$(2k - x)^2 = 4k^2 - 4kx + x^2$$

B1

$$\Rightarrow 2x^2 + 4k^2 - 4kx - 4x - 14 = 0$$

$$(\Rightarrow x^2 + 2k^2 - 2kx - 2x - 7 = 0)$$

$$\Rightarrow x^2 - 2(k+1)x + 2k^2 - 7 = 0$$

AG (be convinced about algebra and = 0)

A1

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3

(ii) $4(k+1)^2 - 4(2k^2 - 7)$

" $b^2 - 4ac$ " in terms of k (either term correct)

M1

$$4k^2 - 8k - 32 = 0 \text{ or } k^2 - 2k - 8 = 0$$

$b^2 - 4ac = 0$ correct quadratic equation in k

A1

$$(k-4)(k+2) = 0$$

Attempt to factorise, solve equation

m1

$$k = -2, k = 4$$

SC B1, B1 for -2 , 4 (if M0 scored)

A1

4

- (iii) Line is a tangent to the circle
Line touches circle at one point etc

E1

1

[17]

Q24.

(a) $(x - 6)^2 + (y - 3)^2$

B1

$= 36 + 9 - 20$

Generous with sign errors

M1

$= 5^2$

Condone 25

A1

3

- (b) (i) Centre (6,3)
ft their a and b

B1 ✓

- (ii) Radius = 5

Correct or ft $\sqrt{RHS}$ if $RHS > 0$

B1 ✓

2

- (c) (i) $x^2 + (x + 4)^2 - 12x - 6(x + 4) + 20 = 0$
Or their $(x - a)^2 + (x + 4 - b)^2 = r^2$

M1

$(2x^2 - 10x + 12 = 0) \Rightarrow x^2 - 5x + 6 = 0$

AG (be convinced)

A1

2

- (ii) $(x - 3)(x - 2) = 0$

Attempt at factors or use of formula

M1

$$x = 2, \quad x = 3$$

Both correct

A1

Substituting for one y value

m1

P, Q are (2,6) and (3,7)

Both points correct

A1

4

[11]

Q25.

(a) $(x - 2)^2 + (y + 1)^2$

$$(x \pm a)^2 + (y \pm b)^2$$

M1

$$= 5^2 \quad \text{or} \quad 25$$

Correct equation for circle

B1A1

3

(b) Sub $x = 6, y = 2$ into *their* circle equation

Or distance $PC^2 = 4^2 + 3^2$

M1

$$(6 - 2)^2 + (2 + 1)^2 = 16 + 9 = 25$$

Shown to equal radius²

A1

2

(c) Gradient $CP = (2 - -1) / (6 - 2)$

Must be y on top and subtraction (6 -2)

M1

$$= \frac{3}{4} = 0.75$$

Any correct equivalent

A1
2

- (d) (i) Grad of perp = -1 / *their* gradient CP
Or $m_1 m_2 = -1$ used or stated

M1

$$= -4/3$$

A1
2

- (ii) $y - 2 =$ their (d)(i) gradient $(x - 6)$
OE such as $3y + 4x = 30$

B1ft
1

[10]



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